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Ionic Liquid and Sulfuric Acid-Based Pretreatment of Bamboo: Biomass Delignification and Enzymatic Hydrolysis for the Production of Reducing Sugars

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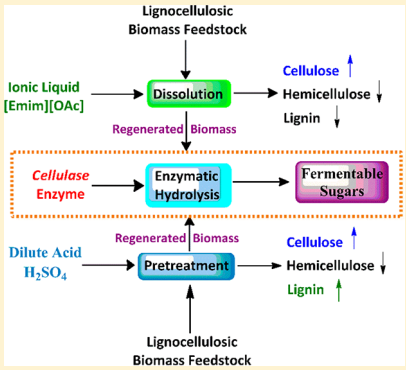
ABSTRACT: The present work investigates the efficiency of two pretreatment pathways of biomass, namely ionic liquid (IL) and dilute sulfuric acid (H_2SO_4) hydrolysis. Both processes are compared in terms of their composition and enzymatic saccharification efficacy. For the IL process, bamboo was dissolved in 1-ethyl-3-methylimidazolium acetate ([Emim][OAc]) at different temperatures (90, 110, 130, and 150 °C) for 3 h. These pretreated bamboo samples were then characterized by thermogravimetric analysis and X-ray diffraction to evaluate the biomass crystallinity and thermal decomposition temperature. The crystallinity index, thermal decomposition temperature, hemicellulose, and lignin content of bamboo were found to decrease after [Emim][OAc] pretreatment. Further, the IL pretreated biomass significantly enhanced the enzymatic saccharification of cellulose component of bamboo. The enzymatic hydrolysis rate for IL pretreated biomass was 4.7 times higher than that of the acid pretreated biomass. This was primarily attributed to the difference in the crystallinity and delignification in the IL process. To improve the enzymatic hydrolysis efficiency of bamboo, a combined pretreatment (dilute acid + ionic liquid) process was also employed and compared with IL pretreated cellulose and bamboo samples. The consequences of this investigation revealed that IL pretreatment may offer novel favorable circumstances compared to a dilute acid pretreatment process for bamboo which can deliver high sugar yields with IL pretreatment.



■ INTRODUCTION

With the predictable diminution of petroleum-based resources, there has been extensive research undertaken to find alternative resources, especially from the sustainable resources containing lignocellulosic biomass.[1,2] Lignocellulosic biomass could be utilized as an alternative renewable source for the production of energy and chemicals for the future due to their availability in huge amounts at a lower cost.[3,4] Depending on the biomass source, it is mainly composed of cellulose (30−50 wt %), hemicellulose (20−35 wt %), lignin (20−30 wt %), and 2−6 wt % of other compounds.[5] For better enzymatic hydrolysis, the pretreatment of lignocellulosic biomass is an important step to enhance the susceptibility of biomass to produce the reducing sugars.[1,6] During the biomass pretreatment process, loss of sugars and inability to decrystallize the cellulose structure results in a costly release of sugars from the biomass.[6,7] Because of the presence of lignin, cellulose crystallinity, and the incidence of extensive covalent cross-linkages between lignin and hemicellulose in the plant cell wall, it is difficult to introduce an effective pretreatment process for biomass recalcitrance.[8] These factors restrict the effective enzymatic hydrolysis of lignocellulosic biomass to produce fermentable reducing sugars in higher amounts.[9,10]

For the above-mentioned reasons, an efficient and novel pretreatment process should have the following features: (a) the ability to disturb the covalent cross-linkages of hemicellulose and lignin which surrounds the cellulose fibers; (b) the ability to interrupt the strong intra- and intermolecular hydrogen linkages present in crystalline cellulose, and (c) help to increase the surface area and porosity of cellulose for efficient enzymatic hydrolysis.[9,11,12] Therefore, several biomass pretreatment processes such as dilute acid, ammonia fiber expansion, hot water, lime, and organic solvents are already employed to overcome the recalcitrance of lignocellulosic biomass, resulting in increased efficiency of enzymatic hydrolysis with an improvement in monomeric sugars yield.[1,13−15] Among all the methods, the dilute sulfuric acid pretreatment method has developed as a prominent pretreatment process for biomass conversion which is now in commercial development.[15] In a dilute sulfuric acid pretreatment process, first hemicellulose is solubilized and thereby disrupts the biomass structure. Nevertheless, it causes the formation of degraded products from polysaccharides and that often provides an inhibitory effect to downstream fermentation organisms which invariably lowers the bioethanol yield.[5,16,17]

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Moreover, the presence of lignin fragments in the biomass after acid pretreatment can also possibly restrict the accessibility of enzymes to the substrate for the production of sugars.[7,18]

While several alternative pretreatment techniques being investigated, novel and effective process technologies are required for the effective conversion of lignocellulosic biomass.[19,20] Ionic liquids (ILs) are the special class of solvents for the dissolution of biomass.[11,19,21,22] Over the past few decades, ILs have already established their potential as a promising solvent for the dissolution of biomacromolecules with high efficiency.[23−27] Recent studies have also reported the comprehensive dissolution and partial delignification of hardwood and softwood biomasses with various ILs.[8,12] Dissolution of biomass using ILs is also based on the interruption of strong inter- and intramolecular hydrogen bonding and formation of new hydrogen bonds between hydroxyl protons and the the anion of the IL.[3,28] The carboxylic acid anion-based ILs are reported to have higher hydrogen bonding acceptor abilities, low melting temperatures, and lesser viscosities. All the above-mentioned properties together easily facilitate the dissolution of biomacromolecules.[29−31]

Among the different biomass wastes, bamboo is an ancient woody grass and scattered in tropical, subtropical, and mellow temperature regions. For the most part, bamboo is viewed as the "poor man's tree,". Over the years, bamboo has been claimed as an innovative, modern raw material and a replacement for wood. According to the State of Forest Report 2011, the total area under bamboo is 13.96 million hectares (8.96 million hectares estimated in 2001). It is found in almost all parts of India from tropical to the temperature regions. In India, bamboo woods contain 125 indigenous and 11 intriguing types of bamboos having a place with 23 genera. The source of bamboo species occurring in India is *Chimonobambusa, Dendrocalamus, Bambusa, Arundinaria, Dinochola, Gigantochloa* etc. More than 50% of the bamboo species occur in Eastern India, namely, Assam, Manipur, Meghalaya, Mizoram, Nagaland, Arunachal Pradesh, Tripura, Sikkim, and West Bengal. *Bambusa and Dendrocalamus* are found under tropical conditions, whereas *Arundinaria* and its allies occur in the temperature region.[32] After China, India, is the second most extravagant nation in bamboo genetic resources. Universally, these two countries have together contributed more than 50 percent of bamboo resources.[33]

Overall, a dilute sulfuric acid pretreatment process can be used to produce reducing sugars from biomass. However, it leads to the formation of higher degradation products from sugars. Therefore, in this work, we have attempted to produce reducing sugars from biomass using the IL pretreatment process with the minimum inhibitory product and also to reduce the time of enzymatic saccharification. The present study attempts to utilize 1-ethyl-3-methylimidazolium acetate ([Emim][OAc]) IL for the dissolution of bamboo biomass. Further, the residual biomass is used for the production of fermentable sugars by enzymatic hydrolysis. Thereafter, the dilute sulfuric acid pretreatment process is also employed for the biomass hydrolysis at 121 °C and different acid concentrations for 2 h. Both IL and dilute acid pretreatments are studied for the resultant saccharification and are evaluated for their feasibility of downstream processing for bioethanol production. The regenerated biomass has been further characterized for glucan, xylan, and lignin content from both processes. The characterization of regenerated biomass has been performed using thermogravimetric analysis (TGA) and X-ray diffraction (XRD) to understand the thermal decomposition behavior and crystallinity of biomass.

■ MATERIALS AND METHODS

Materials. In this study, bamboo biomass (species name: *Bambusa cacharensis* R. B. Majumdar) used as a feedstock which was acquired from IIT Guwahati forestry area, Assam, India. The dried out biomass chips were severed and sieved (mesh size, BSS 30) to obtain a uniform particle size of biomass of 0.5 mm. Further, a comprehensive extraction process for bamboo was performed by means of following the Laboratory Analytical Procedure (LAP) reported by National Renewable Energy Laboratory (NREL), Golden, Colorado, USA. The extracted biomass powder was then dried at 45 °C for 48 h until the residual solvent was evaporated. The obtained dried samples were kept at room temperature in an airtight plastic bag until further use. Microcrystalline cellulose (MCC, DP ≤ 400) powder and sulfuric acid (≥97%) were purchased from Merck, India Pvt. Ltd. Ionic liquid [Emim]-[OAc] (≥95%) and the enzyme *Cellulase* from Trichoderma reessi ATCC 26921 were purchased from Sigma-Aldrich, Germany. The sugars glucose (≥99.5%), xylose (≥99%), cellobiose (≥98%), and arabinose (≥98%), which were used as standards for high-performance liquid chromatography (HPLC) analysis were procured from Sigma-Aldrich, Germany.

Ionic Liquid and Dilute Sulfuric Acid Pretreatment. In a typical dissolution experiment, 10 g of [Emim][OAc] was taken in a 25 mL conical flask. It was dipped in an oil bath where the temperature of the system was kept between 90 to 150 °C with an accuracy of ±1 K. After attaining the desired temperature, 0.5 g of bamboo powder was added to the solvent. The temperature inside the flask was then measured with the help of a digital thermometer, and the mixture was stirred continuously for 3 h. After completion of the reaction time, the residual biomass was regenerated by the addition of distilled water into the solution with 30 min of vigorous stirring. The biomass was collected by using a vacuum filtration unit. Again the biomass residue was washed with an excess amount of hot water to confirm the complete removal of IL. Further, to confirm the complete removal of IL from the pores of bamboo, we have characterized the regenerated biomass by TGA. The percentage solubility of bamboo biomass and individual biomass components are calculated according to the following equation.[3]

$$\text{biomass solubility (\%)} = \frac{m_b - m_{bf}}{m_b} \times 100 \quad (1)$$

$$\text{solubility of cellulose/hemicellulose/lignin (\%)} = \frac{m_o - m_f}{m_o} \times 100 \quad (2)$$

where m_b is the initial mass of bamboo added and m_{bf} is the final mass of recovered bamboo, respectively. m_o and m_f are the initial and final mass of cellulose/hemicellulose/lignin in bamboo, respectively.

Similarly, the pretreatment of bamboo using dilute sulfuric acid was carried out in an autoclave at 1:20 (w/v) solid to liquid ratio and 121 °C for 2 h with 0.2, 0.4, and 0.6 M H_2SO_4. After the reaction, samples were withdrawn from hydrolysate for the estimation of reducing sugars via HPLC. The solid and

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liquid portions were separated by 0.2 μm nylon membrane filter. The solid residue was then washed with distilled water until a pH of 7 was obtained. For both the pretreatment processes, the regenerated biomass samples were dried at 50 °C for 48 h before their compositional analysis. Further, the dried biomass samples were characterized by XRD and TGA. The detailed description of XRD and TGA characterizations have been given in our earlier communications.[1,3]

The pretreatment of bamboo in IL and dilute sulfuric acid was carried out at different temperatures (i.e., 121 °C for acid pretreatment and 90−150 °C for IL pretreatment) and time (3 and 2 h). The selection of temperature and time are based on the recovery of reducing sugars and inhibitory product formations. For acid pretreatment, we have already optimized the pretreatment process with a variety of biomass samples, hence a similar condition has been used in the present study to avoid duplication of data.[34,35] Apart from that, the literature data also suggested the similar operating conditions.[7] At 121 °C and 2 h of dilute acid pretreatment, the removal percentage of hemicellulose was higher with a minimal concentration of inhibitory products. The results of the study also revealed a higher formation of inhibitory products under elevated operating conditions. Therefore, the dilute acid pretreatment was limited to 2 h of reaction time, whereas, in the case of IL pretreatment, 3 h of pretreatment time released maximum sugars at lower temperature and a moderate acid-free environment. It also produced a negligible amount of inhibitory products. Moreover, we made an attempt to extend the study for a prolonged period but going beyond 2.6 h results in the higher viscosity of solution. Therefore, the IL pretreatment was restricted to 3 h. The recovery using ionic liquids even though it required lower operating conditions is recommended as compared to a harsh acidic environment. Our aim hence is to compare both the processes in terms of yield and economic and practical feasibility. It is also a known fact that performance with different pretreatments varies with biomass species. Further, the mechanism of toxin formation with acid and the ionic liquid is not known either detected in our work. To predict the same, one needs to know the cause the changes in substrate features or the way it can disrupt the lignin-carbohydrate linkages. Thereafter, one can confirm the impact on subsequent enzymatic hydrolysis.

Enzymatic Hydrolysis. Batch enzymatic hydrolysis of IL (T = 110 °C, 130 °C, and 150 °C) and dilute acid (0.2 M acid, T = 121 °C) pretreated bamboo samples were carried out in a retaliating shaker at 50 °C and 200 rpm for 72 h. All the biomass samples were diluted in a 50 mM sodium citrate buffer with a pH of 4.8 for enzymatic hydrolysis. The total batch volume for enzymatic hydrolysis was 3 mL with cellulase (ATCC 26921) enzyme concentration of 50 mg of protein per gram of glucan. As reported by NREL, the older filter paper unit (FPU) assay is no longer considered as a meaningful option for the determination of *cellulase* activity because it only converts 3.6% of the substrate and the substrate is not a real-world feedstock.[36] Additionally, NREL measured protein content of *cellulase* and was investigated on the model cellulosic substrate. Therefore, in the present study, instead of measuring the *cellulase* activity in terms of FPU mL^{-1}, the protein content of *cellulase* (Celluclast 1.5 L) was measured according to the Lowry method and loaded in the enzymatic hydrolysis medium;[37] 0.02% of sodium azide was added to preclude the growth of microorganisms.

According to the established NREL procedure (NREL/TP-5100-63351), 20 mg of cellulase protein is recommended for the hydrolysis of a gram of glucan. It was stated that the cellulase blend is produced by the hypercellulolytic mutant *Hypocrea jecorina* (anamorph *Trichoderma reesei*; Cel7A Cellobiohydrolase I).[36] In the present study, we have utilized the cellulase enzyme (derived from *Trichoderma reesei* ATCC 26921; Celluclast 1.5L) for the hydrolysis of glucan. Owing to the presence of a lower protein content in the Celluclast 1.5L, 50 mg of proteins has been loaded for enzymatic hydrolysis. However, according to the economic perspective, the cost of the Celluclast 1.5L is ∼50 times lower than that of Cel7A Cellobiohydrolase I. Therefore, even further usage of 50 mg of protein per gram of glucan does not show much effect on the economics of bioethanol production from lignocellulosic biomass.

In enzymatic hydrolysis, 20 μL of supernatant was taken at a precise time intervals into the 1.5 mL centrifuge tube containing 180 μL of Millipore water, followed by holding the centrifuge tubes in boiling water for 5 min to denature the cellulase. Thereafter, samples were again centrifuged for 5 min and then the supernatant was analyzed to quantify the reducing sugars using HPLC (PerkinElmer Series 200, USA). A Hi-Plex H column (7.7 × 300 mm) was used for HPLC analysis and connected to a guard column. The HPLC column was operated at an oven temperature of 65 °C with 0.5 mL min^{-1} flow rate. The samples were analyzed with a refractive index (RI) detector. H_2SO_4 (0.01 M) was used as a mobile phase. The recognition of peaks was confirmed by standard calibration plots, and the regression coefficient (R^2) was 0.99. The obtained yields of glucose and total reducing sugars were determined according to the following equation.[12,38,39]

$$\text{glucose yield (\%)} = \frac{C_{fg} \times 0.9 \times V}{C_{ig}} \times 100 \quad (3)$$

$$\text{TRS yield (\%)} = \frac{C_{fs} \times V}{C_o} \times 100 \quad (4)$$

where C_{fg} is the glucose concentration after enzymatic hydrolysis (mg·mL^{-1}), C_{ig} is the total concentration of glucose present in bamboo/cellulose after IL or acid pretreatment in mg. C_{fs} is the final concentration of sugars (mg·mL^{-1}), C_o is the initial concentration of bamboo/cellulose after IL or acid pretreatment (mg) and V is the obtained volume of hydrolysate (mL). All the experimental analysis was carried out in duplicate, and the average values are reported.

Chemical Composition of Bamboo. The compositional analysis of pretreated and untreated bamboo samples was performed using standard NREL procedures namely LAP-002 and LAP-005.[40,41] The polysaccharides, such as glucan and xylan were broken down into monomeric sugars through two-step sulfuric acid hydrolysis and further the reducing sugars were quantified by HPLC analysis. Acid-soluble lignin and acid-insoluble lignin were quantified by following the NREL/TP-510-42618 procedure.[40] A UV−vis spectrophotometer was used to measure the acid-soluble lignin (ASL) at an absorbance of 205 nm. The compositional analysis experiments were carried out in triplicate, and the average values were reported.

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RESULTS AND DISCUSSION

Pretreatment of Bamboo Using [Emim][OAc] IL. In our previous study, the COSMO-RS (conductor-like screening model for real solvents) model was employed to screen a total of 1428 ILs comprising 42 cations and 34 anions for the dissolution of cellulose and hemicellulose. On the basis of the logarithmic infinite dilution activity coefficient values (ln γ) and interaction energies of the components in IL, [Emim][OAc] IL was found to be good candidates for the dissolution of cellulose and hemicellulose.[42] Therefore, on the basis of the screening study results and available literature data, [Emim][OAc] was selected for the pretreatment of bamboo.[3,42,43] Further, the ILs [Emim][DEP], [Bmim][OAc], and [Bmim][Cl] are also reported as more effective solvents for the chitin and biomass dissolution. The IL [Emim][OAc] has shown higher solvating power for biomass and its constituents in comparison to other ILs.[7−9,43] [Emim][OAc] IL has beneficial properties such as low toxicity (LD_{50} > 2000 mg kg^{-1}), less viscosity (10 mPa·s at 80 °C), lower melting temperature (−20 °C), low corrosiveness, favorable biodegradability and stability up to 300 °C.[8] The chemical and thermal stability of IL ([Emim][OAc]) used in the present study is relatively low as compared to ammonium and phosphonium-based ILs.[44,45] However, looking at the other advantages, [Emim][OAc] is the best IL to dissolve biomass and its derived compounds. Moreover, [Emim][OAc] can also be recovered and reused. There are several reports which discuss the recovery and reuse of [Emim][OAc].[46,47] Hence, to avoid duplication of data, in the present study we have not attempted the recovery and reuse of IL.

The experimental solubility of bamboo biomass was carried out in [Emim][OAc] at the temperature ranging from 90 to 150 °C for 3 h of reaction time, and the results are reported in Table 1. From the table, it was seen that the solubility of bamboo biomass increases upon an increase in the temperature. This attribution is due to the higher basic nature of the anion in [Emim][OAc] (1.107), which efficiently disrupts both hydrogen bonds (i.e., inter- and intramolecular) present in biopolymers. At 150 °C, 48.27 wt % of bamboo was solubilized in 3 h of reaction time. The remaining biomass that is, the residual biomass, was precipitated by the addition of water and then regenerated with stirring for 30 min. The regenerated residual bamboo was further used for compositional analysis, physical characterizations, and enzymatic hydrolysis.

The dissolution of various types of biomass in different ILs have also been compared with the present study and are reported in Table 1.[48−51] The solubility of bamboo was found to be lower than the triticale straw biomass at 150 °C in [Emim][OAc].[52] As a test case, in the dissolution of southern yellow pine biomass in [Emim][OAC] and [Bmim]Cl ILs, [Emim][OAc] gave greater dissolving ability than [Bmim]Cl at 110 °C for 16 h.[8] This ascription might be because of higher hydrogen bond acceptor ability of $[OAc]^-$ than that of the chloride. It is also a well-known fact that the anion plays a leading role in the dissolution process.[3,53,54]

In our earlier communications, we have proposed few ILs for preferably dissolving cellulose or hemicellulose without impacting the other components of biomass.[2] However, the IL selected in the present study [Emim][OAc] has the capability to dissolve all the three biomass components. The solubility percentage of individual biomass components is measured by using eq 2 and depicted in Figure 1. The

Table 1. Experimental Solubility of Bamboo Biomass in [Emim][OAc] at Different Temperatures and Time along with a Comparison with Other Biomass Dissolution in Ionic Liquids[a,b]

ionic liquid (β)[c]	biomass type	condition (T and t)	solubility (wt %)	refs
[Emim][OAc] (1.107)	bamboo	90 °C, 3 h	21.0	present study
		110 °C, 3 h	31.6	present study
		130 °C, 3 h	42	present study
		150 °C, 3 h	48.3	present study
	wood flour	90 °C, 1.5 h	17.0	Lee et al. (2009)[63]
	southern yellow pine	110 °C, 16 h	98.2	Sun et al. (2009)[8]
	triticale straw	150 °C, 1.5 h	48.8	Fu et al. (2010)[52]
	corn stover	160 °C, 3 h	53.3	Li et al. (2011)[51]
	switch grass	160 °C, 3 h	49.3	Li et al. (2010)[7]
	wheat straw	162 °C, 4.5 h	57.6	Fu and Mazza (2011)[49]
[choline]Gly	rice straw	90 °C, 24 h	48.4	Hou et al. (2012)[48]
[Emim][DEP] (1.00)	cassava pulp	180 °C, 24 h	88.0	Weerachanchai et al. (2012)[50]
[Bmim]Cl (0.84)	southern yellow pine	110 °C, 16 h	52.6	Sun et al. (2009)[8]

[a]Standard uncertainty: $u(T)$ = 0.1 °C, $u(t)$ = 0.05 h, and u(sol) = 1.21 wt % at 95% confidence level. [b]Standard uncertainty calculated by using the following equation,

$$u = \left(\frac{1}{n(n-1)} \sum_{k=1}^{n} (x_{i,k} - X_i)^2 \right)^{1/2}$$

. [c]"β" is the hydrogen bond basicity.

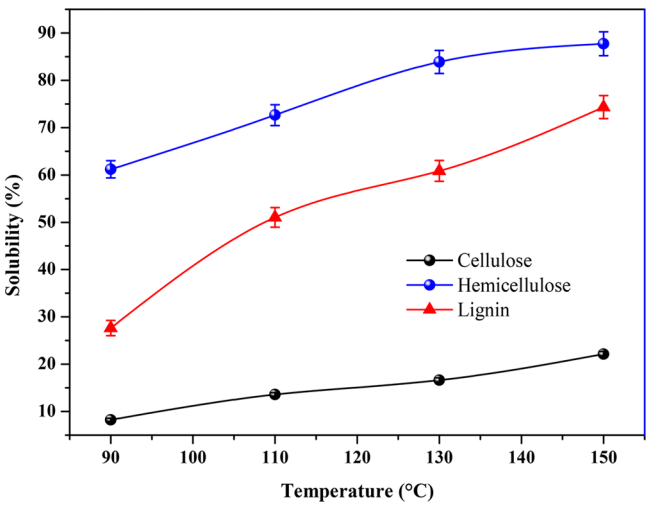

Figure 1. Solubility profiles of bamboo components (cellulose, hemicellulose, and lignin) in [Emim][OAc] at different dissolution temperatures. The linear equation was fitting for solubility (T in °C). Cellulose: $S = 0.23T - 11.66$ (R^2 = 0.99). Hemicellulose: $S = 0.45\ T + 21.88$ (R^2 = 0.96). Lignin: $S = 0.75T - 36.52$ (R^2 = 0.97), respectively.

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solubility of biomass constituents increased with an increase in the temperature. Hemicellulose exhibits an amorphous nature and showed higher solubility in [Emim][OAc] then followed by lignin and cellulose (Figure 1). For the dissolution of biomass in ILs, the anion $[OAc]^-$ forms strong hydrogen bonds with both hemicellulose and lignin thereby increasing their solubility. On the other hand, the anion also forms stronger hydrogen bonds with cellulose but results in lower solubility (Figure 1). This ascription may be because the anion weakens the strong intramolecular and intermolecular hydrogen bonds of cellulose thereby altering the structure of cellulose which results in lower crystallinity of cellulose. In summary, the IL pretreatment removes both lignin (26.55% to 13.16%) and hemicellulose (19.95% to 4.90%) in higher amounts at 150 °C.

Compositional Analysis for Ionic Liquid and Dilute Acid Pretreated Bamboo Biomass. Bamboo was pretreated with 0.2, 0.4, and 0.6 M concentrations of dilute H_2SO_4 at 121 °C for 2 h. After IL and dilute acid pretreatments, the regenerated residual biomass was used for compositional analysis. The compositional analyses of untreated, IL pretreated and dilute acid pretreated bamboo at various experimental conditions are summarized in Table 2. Results of the study show that cellulose content in IL pretreated biomass increases, whereas the lignin and hemicellulose content tends to decrease after IL pretreatment. In the residual biomass, cellulose content (cellulose-rich material) was found to increase from 45.58% to 68.59% at 150 °C for 3 h reaction time, while the IL pretreatment removes both lignin (26.55% to 13.16%) and hemicellulose (19.95% to 4.90%) in higher amounts at 150 °C. The acid pretreatment removes the hemicellulose very effectively and enhances the cellulose and lignin content in comparison to the IL pretreatment process. The hemicellulose content was found to decrease from 19.95% to 1.12% at 0.6 M concentration of dilute sulfuric acid. Overall, the acid pretreatment enhances the cellulose and lignin content from 45.48% to 63.78% and 26.55% to 30.48% at 0.2 M concentration of dilute sulfuric acid.

The recovered bamboo samples have shown substantial reduction of the lignin content in IL pretreatment as compared to the untreated and acid pretreated bamboo. In comparison, IL pretreatment removed 8.47%, 28.41%, 32.58%, and 50.43% of total lignin with 5.58%, 17.93%, 20.02%, and 24.91% of acid insoluble (Klason) lignin and 2.89%, 10.48%, 12.56%, and 25.52% of acid-soluble lignin at 90 °C, 110 °C, 130 °C, and 150 °C, respectively. During the dilute acid pretreatment, lignin content was found to be increased to 14.80%, 9.90%, and −0.67% for 0.2, 0.4, and 0.6 M acid concentrations, respectively. These outcomes showed that IL pretreatment results in a substantial level of lignin reduction.

Sun et al. (2009) reported the dissolution of southern yellow pine and red oak biomass in [Emim][OAc] at 110 °C for 16 h and results in the delignification of 26.1% for southern yellow pine and 34.9% for red oak, respectively.[8] In a similar manner, Li et al. (2010)[7] performed the dissolution of switchgrass in [Emim][OAc] at 160 °C for 3 h, and achieved the lignin reduction of 37.6%, which was less than the delignification efficacy of 50.43% at 150 °C for 3 h in this present study. The difference in the delignification effectiveness might be because of the accompanying reasons: (1) the pretreatments processes generally utilize the higher temperatures and times, resulting in the effective solubilization of lignin, (2) the interactions between biomass and ILs are subject to the ions of IL,

Table 2. Compositional Analysis of Untreated and Pretreated (Ionic Liquid and Dilute Acid) Bamboo Biomass[a],

entry	recovered biomass (%)	glucan (%)	xylan (%)	arabinose (%)	total lignin (%)[b]	AIL (%)	ASL (%)	ash (%)
				Untreated				
	100 ± 0.0 (300 mg)	45.6 ± 0.8 (136.7)	18.2 ± 0.1 (54.7)	1.7 ± 0.3 (5.2)	26.5 ± 0.9 (79.6)	20.0 ± 0.6 (60.0)	06.5 ± 0.3 (19.6)	4.6 ± 0.3 (13.9)
				Ionic Liquid Pretreatment				
90 °C	79.0 ± 1.5 (237.1 mg)	52.9 ± 0.6 (125.4)	8.9 ± 1.2 (21.2)	1.1 ± 0.5 (2.7)	24.3 ± 0.8 (57.6)	16.0 ± 0.5 (37.9)	8.3 ± 0.3 (19.7)	4.3 ± 0.2 (10.2)
110 °C	68.3 ± 1.9 (205 mg)	57.6 ± 1.1 (118.1)	7.3 ± 0.3 (15.0)	1.0 ± 0.3 (2.1)	19.0 ± 1.2 (39.0)	12.0 ± 0.9 (24.6)	7.0 ± 0.5 (14.4)	4.1 ± 0.4 (8.5)
130 °C	58.0 ± 2.1 (174 mg)	65.5 ± 0.4 (114.0)	5.0 ± 0.5 (8.8)	0.8 ± 0.3 (1.3)	17.9 ± 1.0 (31.1)	11.0 ± 0.6 (19.1)	6.9 ± 0.4 (12.0)	4.0 ± 0.2 (6.9)
150 °C	51.7 ± 1.3 (155.2 mg)	68.6 ± 0.8 (106.5)	4.3 ± 0.2 (6.7)	0.6 ± 0.2 (0.9)	13.2 ± 0.5 (20.4)	6.5 ± 0.4 (10.1)	6.7 ± 0.1 (10.3)	3.5 ± 0.3 (5.5)
				Dilute Acid Pretreatment				
0.2 M	62.7 ± 1.7 (188.2 mg)	63.8 ± 0.8 (120.1)	2.3 ± 0.4 (4.3)		30.5 ± 1.1 (57.4)	25.2 ± 0.8 (47.4)	5.3 ± 0.3 (9.9)	3.7 ± 0.3 (7.1)
0.4 M	60.8 ± 2.2 (182.4 mg)	67.5 ± 1.2 (123.1)	1.8 ± 0.7 (3.3)		29.2 ± 0.9 (53.2)	23.4 ± 0.7 (42.7)	5.8 ± 0.2 (10.5)	3.1 ± 0.5 (5.7)
0.6 M	58.6 ± 1.1 (175.7 mg)	71.0 ± 1.7 (124.9)	1.1 ± 0.3 (2.0)		26.4 ± 1.2 (46.3)	20.1 ± 0.8 (35.3)	6.3 ± 0.4 (11.0)	2.4 ± 0.2 (4.3)
				Combined Pretreatment				
	54.7 ± 0.8 (164.2 mg)	81.1 ± 0.9 (133.1)			13.2 ± 0.8 (21.6)	8.5 ± 0.5 (13.9)	4.7 ± 0.3 (7.7)	2.9 ± 0.4 (4.7)

[a]The inability to close the initial mass balance is likely due to the protein and other sugars present in the biomass. Values in parentheses represent the amount of each component recovered from 300 mg total bamboo. cellulose = glucan; hemicellulose = xylan + arabinose. [b]Total lignin = AIL (acid-insoluble lignin (Klason lignin)) + ASL (acid-soluble lignin).

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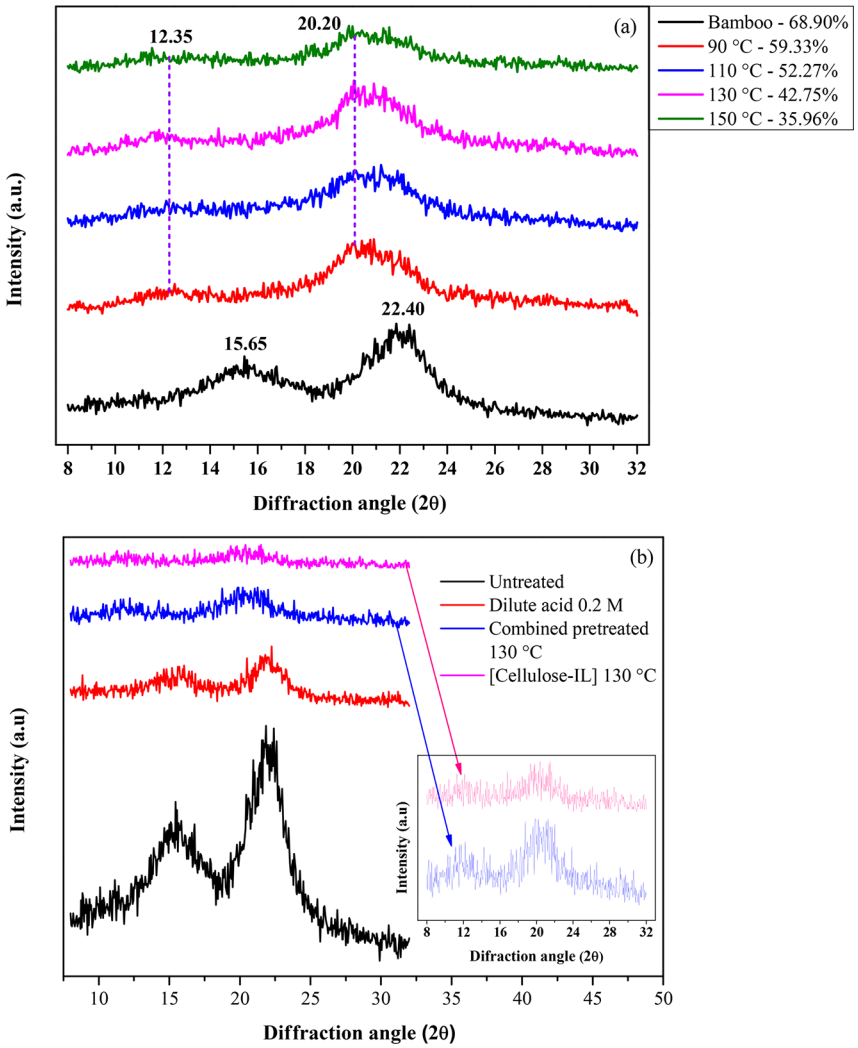

Figure 2. XRD patterns of untreated and (a) pretreated (ionic liquid and dilute acid) bamboo biomass, (b) dilute acid and IL pretreated bamboo/cellulose and combined pretreated bamboo at 130 °C.

temperature, and time utilized as a part of the pretreatment procedure. (3) Lastly, the amount and degree of biomass recalcitrance differs as a component of the biomass itself (i.e., hardwood, softwood, and grass), and is affected by intrinsic variety such as age, harvest methodology, the extent of drying, and storage conditions.

Characterization of Residual Bamboo. *X-ray Diffraction (XRD) Analysis.* The crystallinity index analysis of IL (different temperatures) and dilute acid (0.2 M) pretreated bamboo samples were performed by XRD and compared with the corresponding untreated bamboo sample. The crystallinity of biomass is a critical characteristic influencing parameter in the enzymatic hydrolysis of cellulose. The XRD patterns and crystallinity index values of IL and acid pretreated bamboo samples are presented in Figure 2 and Table 3, respectively. It was reported that the diffraction peak at 15.6° corresponds to the (1 1 0) plane, whereas the second largest peak at 22.4° corresponds to (0 2 0) plane of cellulose I.[55] After dilute acid pretreatment, the XRD peaks are attained at 15.6° and 22.4° which implies that the recovered biomass shows cellulose I structure. From the observation of Table 3, the crystallinity index of 0.2 M acid pretreated bamboo sample was higher (75% CrI) than that of untreated bamboo (68.90% CrI). The increase in the crystallinity of bamboo after acid pretreated sample suggests that the amorphous region of cellulose was broken down under acidic condition. The dilute acid pretreatment method is essential to alter inter- and intramolecular hydrogen bonding present in the cellulose. For acid pretreatment, minimal removal of lignin and higher crystallinity of cellulose are reliable with the results reported in the literature.[7]

Table 3. Crystallinity Index (CrI) and Thermal Decomposition Temperature (T_{dec}) of Pretreated and Untreated Bamboo Samples[a]

pretreatment method	CrI (%)	T_{dec}(°C)
untreated	68.9	363.8
dilute acid (0.2 M)	75.0	361.3
ionic liquid (90 °C)	59.3	354.5
ionic liquid (110 °C)	52.3	352.1
ionic liquid (130 °C)	42.7	346.4
ionic liquid (150 °C)	36.0	344.4
Combined pretreatment	32.4	346.1
Cellulose-IL (130 °C)	20.0	342.3

[a]Standard uncertainty: $u(T)$ = 0.1 °C, and u(CrI) = 0.5%, and $u(T_{dec})$ = 1.2 °C at 95% confidence level

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It was worth noting that the lower crystallinity of biomass is essential to enhance the productivity of reducing sugars during enzymatic hydrolysis. In the case of the IL pretreated bamboo samples, the XRD peaks at 15.6° and 22.4° (native cellulose) diffraction patterns are significantly weakened and shifted to 12.3° and 20.2° (Figure 2a).[3,56] This transformation indicates a change from cellulose I to cellulose II, which occurs during the dissolution of the biomass in ILs.[43] The crystallinity index of the IL pretreated bamboo sample is substantially less than that of the acid pretreated and untreated bamboo samples. The crystallinity of cellulose was also observed to decrease with increasing IL pretreatment temperature (Table 3). This lower crystallinity of cellulose reveals that the regenerated bamboo samples are amorphous in nature and subsequently have higher cellulose surface accessibility and would be more effective for enzymatic hydrolysis.

Thermogravimetric Analysis (TGA). Furthermore, TGA analysis was also carried out on a NETZSCH instrument (TG 209 *F1 Libra*, Germany) under inert atmosphere (N_2). The mass of the sample was about 8−10 mg and loaded into a platinum pan. The TGA instrument was operated at a heating rate of 10 °C min^{-1} over the temperature ranging from 30 to 600 °C. Appraisal of cellulose, hemicellulose, and lignin content of biomass could shed perceptions on the mechanisms contributing to IL and dilute acid pretreatment techniques. The thermal decomposition of lignocellulosic biomass took place in four major steps: (step 1, up to 100 °C) loss of moisture and volatile matter.[57] (step 2, between 100 to 200 °C) the decomposition of residual hemicellulose;[58] (step 3, the peaks at 200 to 400 °C) ascribed to the decomposition of cellulose, hemicellulose, and lignin;[56,58] and (step 4, between 400 to 700 °C) the peak corresponds to the final degradation of cellulose, lignin, and inorganic compounds.[57−59]

Figure 3 shows the thermal decomposition profiles of regenerated and untreated bamboo samples. From DTG curves (Figure 3), it can be seen that the thermal decomposition of bamboo exhibits two distinct stages for untreated and IL pretreated biomass. In the first stage, decomposition occurred in the temperature range of 200−280 °C which may be due to the decomposition of residual hemicellulose. While stage two corresponds to 270−400 °C and exhibits the devolatilization of inorganic compounds.[60] The second stage mainly accounts for thermal decomposition of cellulose, hemicellulose, and trace amounts of lignin, whereas in the case of an acid pretreated biomass sample, only one peak was obtained in the temperature range of 300−400 °C which mainly corresponds to the decomposition of cellulose and trace amount of hemicellulose and lignin.[58] The absence of a hemicellulose decomposition peak in the acid pretreated biomass is due to the complete removal of hemicellulose during the pretreatment step. The acid pretreated biomass showed higher thermal decomposition temperature than the IL pretreated biomass which occurred due to the strong crystallinity of the biomass (Table 3). With an increase in the temperature of IL pretreated biomass, the thermal stability of recovered biomass was found to decrease. From the DTG curves (Figure 3a), the peak at 363.8 °C corresponds to the untreated sample. Whereas, in the case of IL pretreated biomass at different temperatures such as 90, 110, 130, and 150 °C, DTG peaks were obtained at 354.5, 352.1, 346.4, and 344.4 °C, respectively. The decrease in thermal stability of biomass related to the lower crystallinity of recovered bamboo samples. The obtained results in the present

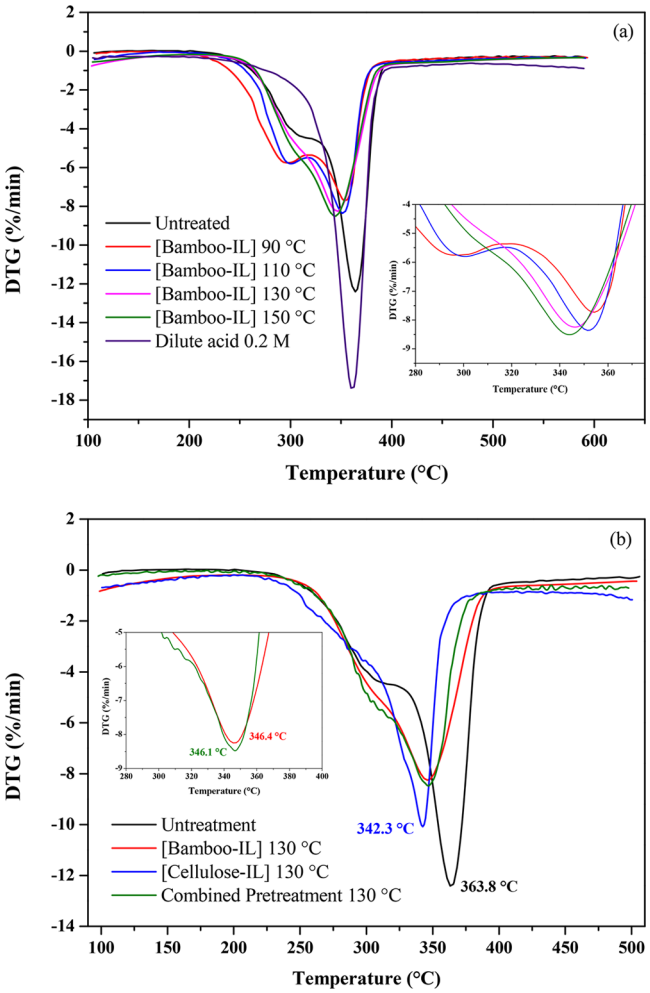

Figure 3. DTG plots of untreated and (a) pretreated (ionic liquid and dilute acid) bamboo biomass, (b) dilute acid and IL pretreated bamboo/cellulose and combined pretreated bamboo at 130 °C.

study showed good agreement with the reported literature.[61,62] It is also worthwhile to mention that, from the TGA analysis, no extra peak was observed in the regenerated biomass which implies that the biomass was successfully regenerated and confirms the removal of IL from the pores of bamboo. After IL and dilute acid pretreatment, the collected residual biomass was further used for enzymatic saccharification.

Enzymatic Saccharification of Ionic Liquid and Dilute Acid Pretreated Biomass. The enzymatic saccharification of both IL and dilute acid pretreated bamboo samples was carried out to produce reducing sugars, and both pretreatment processes were compared. For enzymatic hydrolysis, the IL pretreated bamboo samples at 110, 130, and 150 °C were considered, whereas in the case of dilute acid pretreatment, the sample treated at 121 °C with 0.2 M sulfuric acid was chosen due to the lower degradation of sugars (data not shown). Figure 4a,b shows the total reducing sugars (cellobiose and glucose) and glucose production profiles of IL and dilute acid pretreated bamboo samples at the same enzyme loading (i.e., 50 milligram of protein per gram of glucose). As per the stoichiometry of enzymatic hydrolysis, complete hydrolysis of 1 g of cellulose produces 1.1 g of glucose. Figure 4a shows that the highest TRS yield of 63% was achieved for the 150 °C IL pretreated biomass at 36 h of enzymatic hydrolysis time as compared to 110 °C (40%) and 130 °C (52%) pretreated

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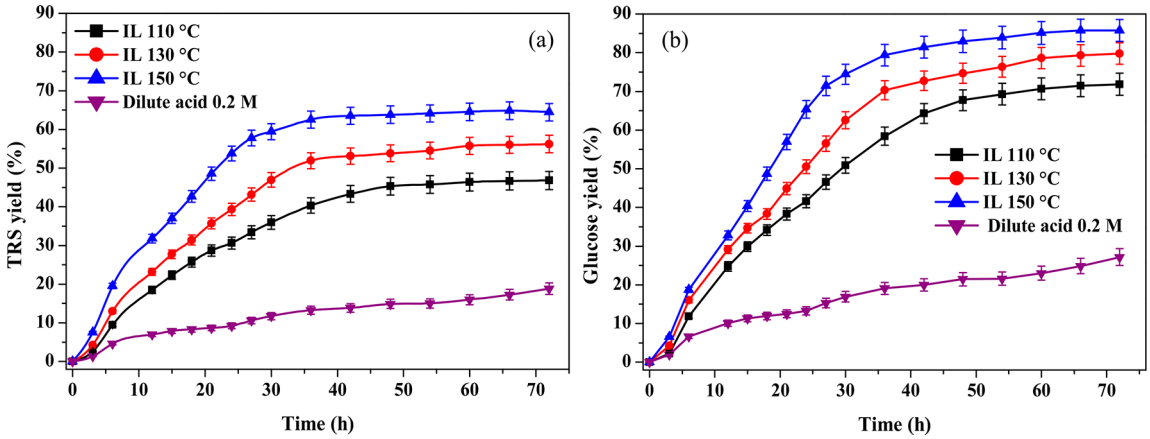

Figure 4. Enzymatic hydrolysis of regenerated bamboo after [Emim][OAc] pretreatment at different temperatures and comparison with dilute acid hydrolyzed biomass (0.2 M H_2SO_4, 121 °C, 2 h) for the production of (a) total reducing sugars (TRS) and (b) glucose.

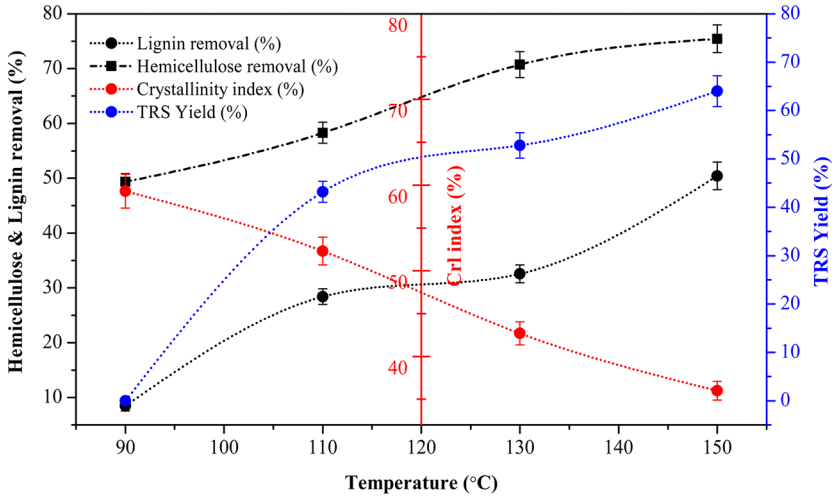

Figure 5. Correlation between the removal percentages of hemicellulose/lignin, the impact of biomass crystallinity, and the TRS yield (after enzymatic hydrolysis at 42 h) obtained from bamboo biomass during IL pretreatment. The yield of TRS for 90 °C treated biomass is considered as zero because the enzymatic hydrolysis was not performed.

biomass. While, in the case of dilute acid pretreated biomass, the highest TRS yield (19%) was obtained at 72 h of enzymatic hydrolysis, which took significantly longer time compared to IL pretreated sample (36 h). Over a similar time interval of the enzymatic hydrolysis process by both pretreatment methods, IL pretreated bamboo exhibited significantly higher enzymatic saccharification than acid pretreated biomass.

The enzymatic hydrolysis rate for IL pretreated biomass was found to be 4.7 times higher than the dilute acid pretreated biomass. This may be ascribed to the difference in the crystallinity and delignification efficiency in IL and acid treated biomass. It was observed that the higher is the lignin content, the lower is the enzymatic saccharification. Dilute acid pretreated biomass revealed significantly higher lignin content in residual biomass as compared to IL pretreated biomass. In the present study, the obtained results suggest a good correlation between the kinetics of enzymatic hydrolysis and effective removal of lignin which is in-line with the observations of Lee et al. (2009)[63] and Li et al. (2010).[12] On the other hand, it was also noticed that the higher the crystallinity index of cellulose, the lower is the reducing sugar production during the enzymatic hydrolysis. Dilute acid pretreated biomass exhibits higher crystallinity index as compared to IL pretreated biomass (Table 3). The loss of intra- and intermolecular hydrogen bonds in cellulose results in amorphous cellulose. This provides an enhanced surface area of cellulose which leads to better accessibility for enzymes and increased active sites in recovered cellulose.

The strong correlations can be seen between lignin/hemicellulose removal, crystallinity index, and TRS yield for IL pretreatment as shown in Figure 5. From the figure, it is clear that with an increase in the dissolution temperature the lignin and hemicellulose removal percentage was found to increase. Simultaneously, the biomass crystallinity was found to decrease. From Figure 5, the TRS yield is increasing with decreasing lignin and hemicellulose content. Higher removal of hemicellulose and lignin components leads to an enhancement in the cellulose content of the residual biomass. The cellulose-rich residual biomass has more likelihood to form new hydrogen bonds with ionic liquids thereby decreasing the crystallinity of biomass. On the other hand, lower lignin content and crystallinity of biomass resulted in higher TRS yield in the enzymatic hydrolysis (see Figure 4 and Figure 5).

Figure 4b represents the glucose yield of IL and the dilute acid pretreated sample. From the observation of Figure 4b, it can be seen that the enzymatic saccharification yield of glucose

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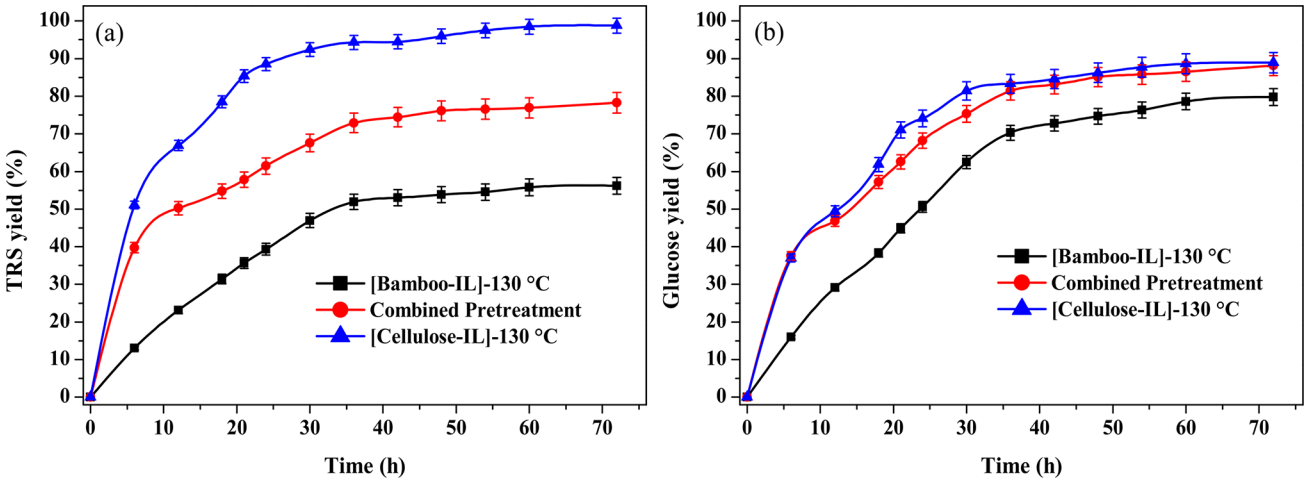

Figure 6. Enzymatic hydrolysis of regenerated bamboo/cellulose after [Emim][OAc] pretreatment at 130 °C for 3 h and comparison with combined pretreated bamboo for the production of (a) total reducing sugars (TRS) and (b) glucose.

was much higher in the IL pretreated biomass (80% for 150 °C at 36 h of enzymatic hydrolysis). Only 27% of glucose yield was achieved in dilute acid pretreated biomass after 72 h of enzymatic saccharification, whereas in the case of IL pretreated biomass, the yield of glucose is also found to be increasing with increase in temperature of IL. For instance at 110 and 130 °C, glucose yield was 58.5% and 70% for 36 h; and 72% and 80% for 72 h, respectively. On the other hand, it was seen that the IL treated biomass has lower thermal stability than untreated and acid treated biomass. From the TGA analysis, it was worthwhile to mention that the release of glucose was higher with decreasing thermal stability of biomass. The decrease in thermal stability of biomass is related to the lower crystallinity of recovered bamboo samples. It was also noticed that the thermal stability of biomass decreased with increasing IL pretreatment temperatures. This, in turn, resulted in higher enzymatic hydrolysis efficiency. However, the amount of produced reducing sugars was very low and required slightly higher operating conditions. Furthermore, to improve the reducing sugars yield, the dilute acid pretreated biomass (reaction condition: 0.2 M, 121 °C, and 2 h) was again treated with [Emim][OAc] at 130 °C for 3 h. Thereafter, the combined pretreated biomass was enzymatically hydrolyzed at the same enzyme loading and the results are compared with IL pretreated sample.

Comparison of Ionic Liquid Pretreated and Combined Pretreated Bamboo. The combined pretreatment process was employed and carried out as a consequence of acid pretreatment followed by IL pretreatment to understand the effect of the combined process on reducing sugars yield. The obtained enzymatic hydrolysis results of IL pretreated bamboo was compared with the combined pretreatment process. After dilute acid pretreatment, the residual biomass was found to contain 64% of cellulose and ∼31% of lignin, respectively. Hence, an attempt has been made to dissolve the dilute acid pretreated biomass in [Emim][OAc], and the study revealed some remarkable results. As can be seen from Table 2, the recovered bamboo contains lower lignin content in the combined pretreated biomass as compared to the IL (130 °C) and acid (0.2 M) pretreated biomasses. This could be due to the presence of fewer components in the residual biomass which provides [Emim][OAc] IL a greater chance for a higher solubilization of lignin at 130 °C. Thus, the combined pretreatment process removed 50.4% of total lignin which results in a higher delignification of recovered biomass at lower pretreatment conditions. The residual biomass recovered at the end of the combined pretreatment process was found to contain 81.07% cellulose which indicates that the recovered biomass is a cellulose-rich material. Apart from that, this study also reported the pretreatment of pure cellulose (microcrystalline cellulose) in [Emim][OAc] at 130 °C for 3 h, and the obtained results are compared with the IL and combined pretreatment processes (Figure 6a,b).

From Figure 6a, the TRS yield was found to be higher in IL pretreated cellulose (94% at 36 h) followed by combined pretreated (73% at 36 h) and IL pretreated (52% at 36 h) bamboo. While in the case of glucose yield (Figure 6b), the concentration of glucose was observed to be practically comparable in both cellulose and combined pretreated bamboo (81−83% at 36 h). The XRD analysis (Table 3 and Figure 2) conferred that the crystallinity index of the combined treated biomass was lower than that of the IL pretreated biomass. This indicates that the recovered biomass is highly amorphous. From Figures 4 and 6, it is also observed that the obtained glucose yield (∼80−81%) was found to be similar in both pretreatment processes, that is, the IL (150 °C) and combined pretreated biomass at 36 h of enzymatic hydrolysis. A further increase in the enzymatic hydrolysis time did not alter the production of glucose. This is due to the enhanced release rate of glucose from the IL pretreated biomass within 36 h. Overall, the enzymatic hydrolysis rate of the IL pretreated biomass is also seen to be higher than that of the acid pretreated biomass due to its lower crystallinity and thermal stability of biomass. This eventually leads to the higher release rate of sugars.

The literature on the hydrolysis of bamboo by ionic liquid pretreatment is rather limited. Therefore, the present study results have been compared with the available literature on bamboo and with the other biomass which has the comparable composition to that of bamboo. Wang et al. (2017) studied the pretreatment of bamboo in three different ILs (1-*H*-3-methylimidazolium chloride [Hmim]Cl, *N*-methyl-2-pyrrolidinium chloride [Hnmp]Cl, and pyridinium chloride [Hpy]Cl) at 90 °C for 0.5 h. The glucose yield reported was 63.5%, 60.8%, and 65.5% after 72 h enzymatic saccharification for [Hpy]Cl, [Hnmp]Cl, and [Hmim]Cl treated biomass samples,

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Table 4. Comparison of Lignin Content and Crystallinity Index (CrI) of Various Biomass at Different Pretreatment Conditions in [Emim][OAc][a]

type of biomass	pretreatment condition (T, t)	lignin (%)		CrI (%)		ref
		untreated	after IL-pretreated	untreated	after IL-pretreated	
bamboo	90 °C, 3 h	26.5	24.3	68.9	59.3	present study
	110 °C, 3 h	26.5	19.0	68.9	52.3	present study
	130 °C, 3 h	26.5	17.9	68.9	42.7	present study
	150 °C, 3 h	26.5	13.2	68.9	36.0	present study
switch grass	160 °C, 3 h	21.8	13.6	26.2	2.6	Li et al. (2010)[7]
corn stover	160 °C, 3 h	20.5	3.5	36.0	N.R[b]	Li et al. (2011)[51]
sugar cane bagasse	120 °C, 0.5 h	24.8	19.8	56.3	24.5	Qiu et al. (2012),[66] Qin and Aita (2013)[67]
agave bagasse	160 °C, 3 h	19.3	10.9	28.6	N.R[b]	Perez-Pimienta et al. (2013)[43]
	120 °C, 3 h	19.3	16.1	28.6	N.R[b]	Perez-Pimienta et al. (2013)[43]
spruce[c,d]	120 °C, 1 h	30.7	28.8	46.7	29.7	Sun et al. (2016)[68]
willow[c,d]	120 °C, 1 h	26.8	24.0	40.1	24.9	Sun et al. (2016)[68]
Soybean straw[c,d]	120 °C, 1 h	17.5	15.4	41.6	23.3	Sun et al. (2016)[68]

[a]Standard uncertainty: $u(T) = 0.1$ °C, u(lignin) = 0.8%, and u(CrI) = 0.5% at 95% confidence level. [b]N.R. = not reported. [c]Combined pretreatment ([Bmim]Cl+Na_2SiO_3 at 1:1 ratio). [d][Bmim]Cl IL was used.

respectively.[64] Whereas, in the present study, 72.2% of glucose yield was obtained after 72 h of enzymatic hydrolysis for 110 °C treated sample. Cheng et al. (2017) also reported the hydrolysis of bamboo in [Bmim]Cl IL with chitosan-based solid acid catalysts. The study revealed a TRS yield of 54.31%, 49.43%, and 58.89% for catalysts Cu^{2+}-SCCR, Zn^{2+}-SCCR, and Fe^{3+}-SCCR, respectively, in [Bmim]Cl at 120 °C and 24 h of dissolution time.[65]

Further, Li et al. (2010)[7] have reported the pretreatment of switchgrass in [Emim][OAc] IL (160 °C for 3 h) followed by its enzymatic saccharification. Almost 85% of glucose yield was observed in 15 h of enzymatic saccharification. As compared to the present study, the crystallinity index of untreated switchgrass (26.2%) was much lower than bamboo (68.9%). During the IL-pretreatment, the crystallinity of switchgrass was reduced to 2.6%, resulting in highly amorphous cellulose. Therefore, there was an increase in the enzymatic hydrolysis efficiency in lesser hydrolysis time. Another study by Lie et al. (2011)[51] investigated on the pretreatment of corn stover using [Emim][OAc] IL at 160 °C for 3 h resulting in 80% cellulose conversion in 24 h by using *cellulase* enzyme. Perez-Pimienta et al. (2013)[43] have reported the pretreatment of agave bagasse in [Emim][OAc] at 160 °C for 3 h; 80% of total reducing sugars were obtained from agave bagasse in 24 h of enzymatic saccharification. However, in the current study, only 65% of TRS was achieved at 36 h (bamboo pretreated at 150 °C in [Emim][OAc]). Qiu et al. (2012)[66] and Qiu and Aita (2013)[67] also studied the pretreatment (120 °C for 30 min) of sugar cane bagasse by an acid catalyzed process in [Emim][OAc]. 80% of glucose yield was obtained after 48 h of enzymatic hydrolysis. In the present study (bamboo pretreated at 130 °C for 3 h), 80% of glucose conversion was achieved within 72 h. In recent years, Sun et al. (2016)[68] studied the effect of combined pretreatment (a method combining solid base catalyst (Na_2SiO_3) and [Bmim]Cl IL) on the spruce, willow, and soybean straw biomass. It was reported that 55−63% of enzymatic hydrolysis yield was obtained after 72 h, which is lower than the present proposed combined pretreatment method.

From the aforementioned discussions, it can be inferred that lignin content and crystallinity index of cellulose plays a vital role in the enzymatic saccharification of any biomass. Hence, the present study results in lower levels of TRS and glucose when compared to those reported in the literature.[43,51,66,67] The diminution in sugars release may depend on the source of biomass, lignin content, and crystallinity of cellulose, pretreatment conditions, enzyme type, and enzyme loading. However, it should be noted that IL pretreated bamboo had lower lignin content and crystallinity index value (Table 4).

The enzymatic hydrolysis results demonstrated that soluble sugars are released at a faster rate in the combined and IL pretreated process than in the dilute acid pretreatment. Moreover, reduction in enzyme loading, ionic liquid cost, and recyclability of ionic liquid are the critical parameters to promote the feasibility of biorefineries and build up the optimum IL pretreatment. Table 5 shows the overall processing time to attain 80% of glucose yield from biomass with different pretreatments. Although the effective acid pretreatment time was 2 h, the total enzymatic saccharification time to attain 80% glucose yield was more than 72 h. In contrast, for IL (150 °C) and combined pretreatments, the total saccharification time to achieve 80% of glucose yield was 36 h. The application of a novel IL pretreatment process offers several advantages for the production of biofuels when compared to the acid pretreatment. The IL pretreatment process is more appropriate and eco-friendly, requires lower processing time for enzymatic hydrolysis, lesser energy consumption, and leads to produce higher sugar yield. It also

Table 5. Total Processing Time (Pretreatment + Enzymatic Hydrolysis) To Recover 80% Yield of Glucan from Various Pretreatment Methods

pretreatment method	pretreatment time (h)	enzymatic saccharification time (h)	total process time (h)
untreated			
dilute acid (0.2 M)	2	≫72	≫74
ionic liquid 130 °C	3	72	∼75
ionic liquid 150 °C	3	37.4	∼40.4
combined pretreatment at 130 °C	2 h acid + 3 h IL	34.4	∼39.4
cellulose-IL 130 °C	3	28.6	∼31.6

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offers lesser degradation of monomeric sugars and consequently the formation of lower inhibitory products for the downstream saccharification step. These advantages need to be counterbalanced by the higher costs of ILs, but offer a better platform to explore and develop this pretreatment technology.

■ CONCLUSIONS

A comparison of IL and acid pretreatment of bamboo was carried at various pretreatment conditions. Among the pretreatment techniques used in this study, IL pretreatment reduces the total enzymatic processing time and provide higher reducing sugar yield. Hence, it is an alternative and promising pretreatment technology to the dilute acid. It was observed that 80% of glucose yield was obtained in the combined pretreatment process while in the case of IL pretreatment (130 °C), 70% of glucose was attained at 36 h of enzymatic hydrolysis. For dilute acid pretreated biomass, only 27% of glucose yield was obtained after 72 h of enzymatic saccharification. The XRD and TGA analysis of the recovered bamboo showed lower crystallinity index value and thermal stability than untreated and dilute acid pretreated biomass. The lower crystallinity of IL pretreated bamboo results in an increase in the cellulose surface accessibility and makes it more easily accessible to enzymatic hydrolysis. With the increase in the IL pretreated temperature, the crystallinity and thermal stability of biomass is found to be lower which results in higher efficiency of enzymatic hydrolysis rate. IL pretreatment is a comparatively new process and there is much to be explored before a commercially reasonable process is comprehended. Moreover, it appears to be environmentally benign and offers numerous advantages over acid pretreatment in the matter of processing time and production of reducing sugars with minimal inhibitory content.

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Notes

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